

Time-Reversed BSDEs for Accurate Gradient Estimation in Diffusion Models

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Abstract—There is a growing literature adopting a stochastic optimal control (SOC) perspective to fine-tune diffusion models and related generative policies. A prominent class of methods, known as *iterative diffusion optimization*, solves the SOC problem by simulating the diffusion process, evaluating a loss function, and applying stochastic optimization algorithms, with *adjoint matching* emerging as a state-of-the-art approach. However, the adjoint process used in these methods is not adapted to the forward diffusion filtration, which can lead to unstable or high-variance gradient estimates. In this paper, we revisit gradient estimation in diffusion models through the lens of backward stochastic differential equations (BSDEs). We propose an alternative estimator based on a time-reversed BSDE formulation introduced in our prior work, which produces an adjoint process adapted to the underlying filtration. This adapted structure leads to more stable gradient estimates with potentially lower variance. We analyze the accuracy of the proposed estimator and compare it with adjoint matching. Numerical experiments on fine-tuning toy diffusion models demonstrate improved gradient stability and competitive performance.

I. INTRODUCTION

Generative diffusion models [1], [2], [3] have demonstrated remarkable capability in generating samples from complex high-dimensional distributions, across a wide range of applications, including images synthesis [4], protein structure generation [5], text generation [6], video generation [7], and robotics control policy generation [8]. In many practical scenarios, however, it is desirable to steer the generation process toward outputs that satisfy additional objectives, preferences, or task-specific requirements. Compared to training a new diffusion model from scratch, fine-tuning an existing model provides a more efficient and cost-effective approach.

A common strategy to fine-tuning diffusion models is to introduce a reward function that measures the quality of generated samples. This formulation naturally enables the use of both reinforcement learning (RL) [9], [10] and stochastic optimal control (SOC) methods [11], [12], [13]. The interplay between SOC and generative modeling has recently garnered significant attention [14], [15], [16], [17], [18], [19]. Motivated by these developments, we adopt the SOC perspective in this paper and focus on a class of SOC-based methods known as *iterative diffusion optimization* [20]. These methods fine-tune the model by repeatedly simulating the diffusion process, evaluating a loss function on the generated samples, and updating the parameters via stochastic optimization. Among them, adjoint matching has emerged

as a state-of-the-art technique [12]. However, the adjoint process employed in these methods is not adapted to the forward diffusion filtration, which may lead to unstable or high-variance gradient estimates.

These limitations motivate the development of more principled adjoint methods in which the adjoint process is adapted to the forward filtration and characterized by a backward stochastic differential equation (BSDE), as introduced in the stochastic maximum principle [21]. In our prior work [22], [23], we introduced a novel time-reversal methodology for solving such BSDEs and showed that it compares favorably with existing BSDE solvers in the linear setting. In this paper, we extend this numerical framework to the nonlinear setting and demonstrate its effectiveness for fine-tuning nonlinear diffusion models.

Here are the contributions of the paper: (i) We propose a novel and accurate gradient estimator for fine-tuning diffusion models; (ii) we present both adapted and non-adapted adjoint processes and establish a theoretical comparison; (iii) We conduct extensive numerical experiments comparing our method with baseline approaches on both linear and nonlinear stochastic models, and demonstrate its effectiveness, outperforming adjoint matching in a full fine-tuning task on a toy diffusion model.

The paper is organized as follows: (i) the SOC problem for fine-tuning diffusion models is introduced in Sec. II; (ii) the non-adapted and adapted adjoint processes are introduced and compared in Sec. III; (iii) the proposed time-reversal methodology to solve the BSDE is presented in Sec. IV; (iv) finally, numerical results comparing the performance of our method with adjoint matching are presented in Sec. V.

II. PROBLEM FORMULATION

Consider a n -dimensional diffusion process $X^o = \{X_t^o; 0 \leq t \leq T\}$ governed by the following stochastic differential equation (SDE)

$$dX_t^o = f(t, X_t^o) dt + g(t, X_t^o) dW_t, \quad X_0^o \sim p_0, \quad (1)$$

where $W = \{W_t; 0 \leq t \leq T\}$ is the n -dimensional standard Wiener process, p_0 is the initial distribution, $f: \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the drift function, and $g: \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$ is the diffusion matrix. These functions are assumed to satisfy the usual regularity conditions that ensure the existence of a strong solution [24, Thm. 5.2.1]. We interpret the SDE (1) as a pre-trained diffusion model [3]. Specifically, the initial distribution, the drift function, and the diffusion matrix are designed such that the distribution of X_T^o , at the terminal time T , matches a target distribution p_T representing data.

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We are interested in the problem of fine-tuning the diffusion model so that it generates samples from the “tilted” target distribution

$$q_T^* \propto p_T e^{-\ell_f},$$

where ℓ_f is a user-specified loss function that evaluates the quality of generated samples. The modified or the controlled diffusion is denoted by $X = \{X_t; 0 \leq t \leq T\}$ and modeled by

$$dX_t = f(t, X_t) dt + g(t, X_t)(U_t dt + dW_t), \quad X_0 \sim q_0, \quad (2)$$

where q_0 is the modified initial distribution, and $U = \{U_t; 0 \leq t \leq T\}$ is the control input process which is assumed to be adapted to the filtration generated by the initial condition and the Wiener process.

The problem of sampling from the tilted target distribution can now be formulated as a stochastic control problem in which the initial distribution q_0 and the control input U are selected so that the distribution of X_T matches the tilted target distribution q_T^* . Using a KL-regularized variational characterization of the tilted distribution, together with Girsanov’s theorem, this problem can be expressed as the following stochastic optimal control problem:

$$\min_{q_0, U} \mathbb{E} \left[\ell_f(X_T) + \int_0^T \frac{1}{2} \|U_t\|^2 dt \right] + D_{\text{KL}}(q_0 \| p_0). \quad (3)$$

Stochastic optimal control problems of this form have a long history in the control literature, particularly in path-integral control [25], [26], [27], where the control-affine structure of the model (2) and the quadratic control cost in (3) arises naturally through a change-of-measure argument related to Girsanov’s theorem. In particular, the optimal controlled diffusion process that solves the SOC problem (3) has the path probability law

$$\frac{dP_X}{dP_{X^o}} = \frac{\exp(-\ell_f(X_T^o))}{\mathbb{E}[\exp(-\ell_f(X_T^o))]},$$

where the expectation is taken with respect to the original diffusion process (1). Specifically, the terminal distribution of the optimal controlled process X_T coincides with the tilted target distribution q_T^* . The path probability law allows us to derive expressions for the optimal initial distribution and the optimal control [11, Thm. 1], however, accurately estimating the expectations is challenging due to the high variance of the exponential weight $\exp(-\ell_f(X_T^o))$ [28], [29]. In this paper, we investigate an alternative class of optimization-based approaches, motivated by recent advances in diffusion models.

III. ADJOINT PROCESSES

In this section, we present two adjoint processes that arise in the gradient estimation of the stochastic optimal control problem (3). The first is a *non-adapted adjoint process* that appears in adjoint matching methods and is obtained via pathwise differentiation of the objective. The second is an *adapted adjoint process* obtained from a BSDE.

A. Notations

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ be a filtered probability space and assume the Wiener process W_t is adapted to the filtration $\mathcal{F}_t$. We will use the following Hilbert spaces throughout the exposition: (i) The Euclidean space $\mathbb{R}^n$ equipped with the inner-product $\langle x, y \rangle := x^\top y$; (ii) The space of deterministic control trajectories $L^2([0, T]; \mathbb{R}^n)$ equipped with the inner product $\langle u, v \rangle := \int_0^T u_t^\top v_t dt$; (iii) The space of $\mathbb{R}^n$ -valued $\mathcal{F}_0$ -measurable random variables $L^2_{\mathcal{F}_0}(\Omega; \mathbb{R}^n)$ equipped with the inner-product $\langle X, Y \rangle := \mathbb{E}[X^\top Y]$; (iv) The space of $\mathcal{F}_t$ -adapted stochastic control processes $L^2_{\mathcal{F}}(\Omega \times [0, T]; \mathbb{R}^n)$ equipped with the inner product $\langle U, V \rangle := \mathbb{E}[\int_0^T U_t^\top V_t dt]$.

For a functional $F : \mathcal{H} \rightarrow \mathbb{R}$ defined on a Hilbert space $\mathcal{H}$ with inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$, the Gâteaux derivative of F at $u \in \mathcal{H}$ is denoted by $\frac{\partial F}{\partial u}(u) \in \mathcal{H}$ and satisfies

$$\frac{d}{dt}(F(u + tv)) = \langle \frac{\partial F}{\partial u}(u), v \rangle_{\mathcal{H}}, \quad \forall v \in \mathcal{H}.$$

For any initial condition $x \in \mathbb{R}^n$ and deterministic control $u \in L^2([0, T]; \mathbb{R}^n)$, we use $X_t^{x, u}$ to denote the solution to the controlled SDE (2) with $X_0^{x, u} = x$ and $U_t = u_t$. Similarly, for any $\mathcal{F}_0$ -measurable random variable $\xi \in L^2_{\mathcal{F}_0}(\Omega; \mathbb{R}^n)$, and admissible stochastic control process $U \in L^2_{\mathcal{F}}([0, T]; \mathbb{R}^n)$, we denote by $X_t^{\xi, U}$ the solution to the same SDE with the random initial condition $X_0^{\xi, U} = \xi$ and the control input U . In particular, the controlled diffusion (2) can be written as $X_t = X_t^{\xi, U}$ where the distribution of ξ is equal to q_0 .

Using this notation, define the sample objective function

$$J(x, u)(\omega) := \beta \ell_f(X_T^{x, u}(\omega)) + \int_0^T \frac{1}{2} \|u_t\|^2 dt, \quad \omega \in \Omega.$$

When the initial condition and control are random, the corresponding objective function is written as

$$J(\xi, U) = \beta \ell_f(X_T^{\xi, U}) + \int_0^T \frac{1}{2} \|U_t\|^2 dt.$$

With this notation, the SOC objective function in (3) can be expressed compactly as

$$\mathbb{E}[J(\xi, U)] + D_{\text{KL}}(q_0 \| p_0), \quad \text{where } \xi \sim q_0.$$

For the purposes of this section we temporarily ignore the KL-divergence term and focus on the derivatives of $\mathbb{E}[J(\xi, U)]$. The derivative of the KL term will be addressed separately in Sec. V-C.

B. Non-adapted adjoint process

Throughout this subsection we assume the diffusion matrix is independent of the state, i.e. $g(t, x) = g(t)$. The non-adapted adjoint process arises when computing the derivatives of the sample objective $J(x, u)(\omega)$ with respect to the initial condition $x \in \mathbb{R}^n$ and deterministic control trajectory $u \in L^2([0, T]; \mathbb{R}^n)$, for a fixed sample $\omega \in \Omega$. Under suitable regularity conditions [30, Ch. V, Thm. 39], the mapping $x \rightarrow X_t^{x, u}(\omega)$ is known to be differentiable and the sensitivity matrix $\eta_t^x(\omega) := \frac{\partial X_t^{x, u}(\omega)}{\partial x} \in \mathbb{R}^{n \times n}$ solves the differential

equation

$$\frac{d}{dt}\eta_t^x = \frac{\partial f}{\partial x}(t, X_t^{x,u})\eta_t^x, \quad \eta_0^x = I.$$

The (non-adapted) adjoint process $Y_t^{x,u}$, is defined by the backward differential equation

$$\frac{dY_t^{x,u}}{dt} = -\left[\frac{\partial f}{\partial x}(t, X_t^{x,u})\right]^\top Y_t^{x,u}, \quad Y_T^{x,u} = \nabla \ell_f(X_T^{x,u}). \quad (4)$$

so that the duality relationship $\frac{d}{dt}((Y_t^{x,u})^\top \eta_t^x) = 0$ holds true. This identity allows the derivatives of the objective to be expressed in terms of the adjoint process:

$$\begin{aligned} \frac{\partial J}{\partial x}(x, u)(\omega) &= Y_0^{x,u}(\omega), \\ \left(\frac{\partial J}{\partial u}(x, u)(\omega)\right)_t &= u_t + g(t)^\top Y_t^{x,u}(\omega). \end{aligned}$$

The derivation follows from a standard variational analysis of the objective function $J(x, u)(\omega)$, for a fixed $\omega \in \Omega$, combined with the above duality relation and is omitted for brevity.

For a random initial condition $\xi \in L^2_{\mathcal{F}_0}(\Omega; \mathbb{R}^n)$ and a stochastic control $U \in L^2_{\mathcal{F}}(\Omega \times [0, T]; \mathbb{R}^n)$, the corresponding adjoint process is denoted by Y_t and formally defined as

$$Y_t(\omega) := Y_t^{\xi(\omega), U(\omega)}(\omega).$$

C. Adapted adjoint process

The adjoint process appearing in the previous subsection is not $\mathcal{F}_t$ -adapted. In contrast, the stochastic maximum principle (SMP) introduces an adapted adjoint process obtained as the solution of a BSDE [21]. Define the Hamiltonian

$$H(t, x, y, z, u) := \frac{1}{2}\|u\|^2 + y^\top (f(t, x) + g(t, x)u) + \text{Tr}(zg(t, x)), \quad (5)$$

for $(t, x, y, z, u) \in [0, T] \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^{n \times n} \times \mathbb{R}^n$. The adapted adjoint process (Y_t, Z_t) is defined by the BSDE

$$-dY_t = \frac{\partial H}{\partial x}(t, X_t, Y_t, Z_t, U_t) dt - Z_t dW_t, \quad Y_T = \nabla \ell_f(X_T). \quad (6)$$

The adaptedness of (Y_t, Z_t) enables a variational analysis of $\mathbb{E}[J(\xi, U)]$ [21, Sec. 4.3]. In particular, for any perturbations $\zeta \in L^2_{\mathcal{F}_0}(\Omega; \mathbb{R}^n)$ and $V \in L^2_{\mathcal{F}}(\Omega \times [0, T]; \mathbb{R}^n)$, we have

$$\begin{aligned} \frac{d}{d\varepsilon} \mathbb{E}[J(\xi + \varepsilon \zeta, U)] \Big|_{\varepsilon=0} &= \mathbb{E}[Y_0^\top \zeta] \\ \frac{d}{d\varepsilon} \mathbb{E}[J(\xi, U + \varepsilon V)] \Big|_{\varepsilon=0} &= \mathbb{E} \left[\int_0^T \frac{\partial H}{\partial u}(t, X_t, Y_t, Z_t, U_t)^\top V_t dt \right]. \end{aligned}$$

Consequently, following the definition of Gâteaux derivative,

$$\begin{aligned} \frac{\partial}{\partial \xi} \mathbb{E}[J(\xi, U)] &= Y_0, \\ \left(\frac{\partial}{\partial U} \mathbb{E}[J(\xi, U)] \right)_t &= U_t + g(t, X_t)^\top Y_t. \end{aligned}$$

This variational analysis differs slightly, and is in fact simpler, than the one used in the SMP. In particular, it considers variations in the Hilbert space $L^2_{\mathcal{F}}(\Omega \times [0, T]; \mathbb{R}^n)$ rather than the needle-type perturbations that are typically employed in the SMP framework.

D. Relationship between adapted and non-adapted adjoint processes

Assume again that $g(t, x) = g(t)$. A relationship between the two adjoint processes is provided using the linearity of the BSDE (6). Let $A_t := \frac{\partial f}{\partial x}(t, X_t)$, and define the matrix processes

$$d\Phi_t = -A_t^\top \Phi_t dt, \quad d\Psi_t = \Psi_t A_t^\top dt, \quad \Phi_0 = \Psi_0 = I.$$

By the application of the Itô rule, one can verify that $\Psi_t \Phi_t = I$, or $\Phi_t = \Psi_t^{-1}$, for all $t \geq 0$. These matrix-valued processes are used to provide an explicit solution to the BSDE [21, Sec. 7.2]. The BSDE dynamics yields $d(\Psi_t Y_t) = \Psi_t Z_t dW_t$. Integrating from t to T , and multiplying by Φ_t , gives

$$Y_t = \Phi_t \Psi_T \nabla \ell_f(X_T) - \int_t^T \Phi_t \Psi_s Z_s dW_s,$$

where we used the terminal condition $Y_T = \nabla \ell_f(X_T)$. The solution Y_t is obtained by taking conditional expectation

$$Y_t = \mathbb{E}[\Phi_t \Psi_T \nabla \ell_f(X_T) \mid \mathcal{F}_t].$$

On the other hand, the non-adapted adjoint satisfies $Y_t = \Phi_t \Psi_T \nabla \ell_f(X_T)$, concluding the identity

$$Y_t = \mathbb{E}[Y_t \mid \mathcal{F}_t].$$

Thus the adapted adjoint process can be interpreted as the conditional expectation of the pathwise adjoint process, implying variance reduction by projection onto the filtration. Moreover, the difference between Y_t and Y_t can be written explicitly as

$$Y_t - Y_t = \int_t^T \Phi_t \Psi_s Z_s dW_s.$$

From a computational standpoint, the two adjoint processes behave differently. The non-adapted adjoint Y_t is obtained by solving a backward ordinary differential equation along each simulated trajectory, which makes it straightforward to compute once the forward trajectory X_t is known. In contrast, the adapted adjoint (Y_t, Z_t) is numerically challenging to simulate because the BSDE (6) imposes a terminal condition at time T , while the filtration $\{\mathcal{F}_t\}$ is generated forward in time. Our proposed methodology deals with this issue.

IV. PROPOSED METHODOLOGY

Our numerical approach is based on a time-reversed formulation of the BSDE (6). This construction relies on two key ingredients: (i) the connection between BSDEs and certain partial differential equation (PDE), and (ii) the time-reversal of diffusion processes. To present this construction, we assume that the control input is given in feedback form,

$$U_t = k(t, X_t). \quad (7)$$

where $k : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a feedback policy. Under this assumption, the controlled diffusion (2) becomes a Markov process whose dynamics are fully determined by the state X_t . This Markov structure allows the solution of the BSDE to be characterized through a PDE representation and enables the derivation of a time-reversed formulation that forms the basis of our numerical algorithm.

A. PDE solution of the BSDE

We begin by recalling the well-known connection between BSDE and PDE [31], [32]. Let $\phi : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a sufficiently smooth function satisfying the terminal value problem

$$\frac{\partial \phi}{\partial t}(t, x) + \mathcal{L}\phi(t, x) + h(t, x) = 0, \quad \phi(T, x) = \nabla \ell_f(x), \quad (8)$$

where the second-order differential operator $\mathcal{L}$ is defined by

$$\begin{aligned} \mathcal{L}\phi(t, x) := & \frac{\partial \phi}{\partial x}(t, x)^\top (f(t, x) + g(t, x)k(t, x)) \\ & + \frac{1}{2} \text{Tr}(G(t, x) \frac{\partial^2 \phi}{\partial x^2}(t, x)), \end{aligned}$$

with $G(t, x) := g(t, x)g(t, x)^\top$, and

$$h(t, x) := \frac{\partial H}{\partial x}(t, x, \phi(t, x), g(t, x)^\top \partial_x \phi(t, x), k(t, x)). \quad (9)$$

Applying Itô's formula to $\phi(t, X_t)$ along the controlled diffusion (2) shows that the processes

$$(Y_t, Z_t) = (\phi(t, X_t), g(t, X_t)^\top \frac{\partial \phi}{\partial x}(t, X_t)) \quad (10)$$

solves the BSDE (6). Throughout the paper we assume that the PDE (8) admits a sufficiently regular solution. Such existence and regularity results hold under standard conditions on the coefficients of the diffusion and the Hamiltonian; see, e.g., [21, Ch. 7 Sec. 4]. While solving the PDE directly is generally intractable in high dimensions, this representation provides the key insight needed to construct a time-reversed stochastic representation of the BSDE.

B. Time-reversed diffusion

Let $\tilde{X} := \{\tilde{X}_t := X_{T-t}; 0 \leq t \leq T\}$ denote the time-reversal of the diffusion process X . According to the time-reversal theory for diffusion processes [33], [34], the time-reversed process $\tilde{X}$ satisfies the SDE

$$d\tilde{X}_t = \tilde{f}(t, \tilde{X}_t) dt + \tilde{g}(t, \tilde{X}_t)(\tilde{U}_t dt + d\tilde{W}_t), \quad (11)$$

where $\tilde{W}_t$ is a standard n -dimensional Wiener process (independent of W_t). The coefficients of the reversed diffusion are given by

$$\begin{aligned} \tilde{f}(t, x) &:= -f(T-t, x) + s(T-t, x), \\ \tilde{g}(t, x) &:= -g(T-t, x), \quad \tilde{U}_t := k(T-t, \tilde{X}_t), \end{aligned}$$

where $s : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ denotes the *score function* associated with the marginal distribution of X_t . The score function is defined component-wise as

$$s_i(t, x) := \frac{1}{p(t, x)} \sum_{j=1}^n \frac{\partial}{\partial x_j} (G_{i,j}(t, x) p(t, x)), \quad (12)$$

where $p(t, x)$ denote the probability density function of X_t , $G_{i,j}(t, x)$ is the (i, j) -entry of the matrix $G(t, x)$.

In practice, the score function is not known explicitly and must be approximated from samples of the forward diffusion

process. A common approach is to estimate the score by solving the *implicit score matching* problem

$$\min_{\Psi} \mathbb{E} \left[\frac{1}{2} \|\Psi(t, X_t)\|^2 + \sum_{i,j=1}^n G_{i,j}(t, X_t) \frac{\partial \Psi_i}{\partial x_j}(t, X_t) \right], \quad (13)$$

where the expectation is taken with respect to both $t \sim \text{Unif}[0, T]$ and the state X_t of the forward diffusion [1], [35]. The time-reversed diffusion will allow us to construct a time-reversed BSDE which is described next.

Algorithm 1 Time-Reversal BSDE solver

- 1: **Input:** sample size N , step-size Δt , iteration J_f , function classes Ψ and Φ , initial distribution q_0 .
 - 2: Initialize $\{X_0^i\}_{i=1}^N \sim q_0$
 - 3: Obtain $\{X_t^i\}_{i=1}^N$ by simulating (2)
 - 4: Learn the score function $s(t, \cdot)$ through the implicit score matching optimization (13)
 - 5: Set $\tilde{X}_0^i = X_T^i$ for $i = 1, \dots, N$.
 - 6: Obtain $\{\tilde{X}_t^i\}_{i=1}^N$ by simulating (11)
 - 7: Initialize the function $\phi \in \Phi$
 - 8: **for** $j \in \{1, 2, \dots, J_f\}$ **do**
 - 9: $\tilde{Y}_0^i = \nabla \ell_f(\tilde{X}_0^i)$
 - 10: Obtain $\{\tilde{Y}_t^i\}_{i=1}^N$ by simulating (15)
 - 11: Update ϕ using ADAM optimizer minimizing (16)
 - 12: **end for**
 - 13: **Output:** $\{\phi(t, \cdot)\}_{t \in \{0, \Delta t, \dots, T\}}$
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C. Time-reversed BSDE

The time-reversed BSDE is obtained by applying Itô's formula to the process $\phi(T-t, \tilde{X}_t)$, where ϕ is the solution of the PDE (8) and $\tilde{X}_t$ evolves according to the time-reversed diffusion (11). After simplification, this yields

$$\begin{aligned} d\phi(T-t, \tilde{X}_t) &= (h(T-t, \tilde{X}_t) + c(T-t, \tilde{X}_t)) dt \\ &\quad + \frac{\partial \phi}{\partial x}(T-t, \tilde{X}_t)^\top \tilde{g}(t, \tilde{X}_t) d\tilde{W}_t \end{aligned}$$

where h is the partial derivative of the Hamiltonian as defined in (9) and

$$c(t, x) := \frac{\partial \phi}{\partial x}(t, x)^\top s(t, x) + \text{Tr}(G(t, x) \frac{\partial^2 \phi}{\partial x^2}(t, x)).$$

Define the processes

$$(\tilde{Y}_t, \tilde{Z}_t) := (\phi(T-t, \tilde{X}_t), \tilde{g}(t, \tilde{X}_t)^\top \frac{\partial \phi}{\partial x}(T-t, \tilde{X}_t)). \quad (14)$$

Then $\tilde{Y}_t$ satisfies the SDE

$$\begin{aligned} d\tilde{Y}_t &= \frac{\partial H}{\partial x}(T-t, \tilde{X}_t, \tilde{Y}_t, \tilde{Z}_t, \tilde{U}_t) dt + c(T-t, \tilde{X}_t) dt + \tilde{Z}_t^\top d\tilde{W}_t, \\ \tilde{Y}_0 &= \nabla \ell_f(\tilde{X}_0), \end{aligned} \quad (15)$$

which can be interpreted as the time-reversal of the BSDE (6). By construction, the path $\{(\tilde{Y}_t, \tilde{Z}_t); 0 \leq t \leq T\}$ has the same probability law as $\{(Y_{T-t}, Z_{T-t}); 0 \leq t \leq T\}$.

D. Numerical procedure

The key advantage of the representation (15) is that, unlike the original BSDE (6), it does not impose a terminal constraint. Instead, the process can be simulated forward in time starting from the initial condition $\tilde{Y}_0 = \nabla \ell_f(\tilde{X}_0)$. This property allows the BSDE to be approximated using standard forward simulation techniques.

The simulation of (15), however, requires the knowledge of ϕ . To overcome this challenge, we follow an iterative procedure, where we start with an initial guess for ϕ , simulate (15), and update ϕ through the optimization

$$\min_{\phi \in \Phi} \sum_{i=1}^N \|\tilde{Y}_t^i - \phi(T-t, \tilde{X}_t^i)\|^2 \quad (16)$$

where $\{(\tilde{X}_t^i, \tilde{Y}_t^i)\}_{i=1}^N$ are simulated trajectories generated by the coupled system (11)–(15). The time-reversed BSDE (15) therefore provides the basis for a practical numerical algorithm for solving the original BSDE. The details appear in Algorithm 1.

Remark 1: The proposed methodology remains applicable even when the score function s is only approximately known. In particular, the function ϕ can still be learned by solving the regression problem (16). In practice, the accuracy of the learned function ϕ is highest in regions of the state space where the reversed trajectories are sampled more frequently. The role of the score function is to ensure that these trajectories remain consistent with the forward diffusion, thereby concentrating samples in regions that are most relevant for the SOC problem and improving the accuracy of ϕ where it matters most for the optimization.

E. Iterative procedure and its convergence analysis

The more detailed iterative procedure and its convergence analysis are presented in this section. Starting from an initialization ϕ^0 , at iteration j the algorithm simulates (15) with ϕ replaced by ϕ^{j-1} :

$$\begin{aligned} d\tilde{Y}_t^j &= \frac{\partial H}{\partial x}(T-t, \tilde{X}_t, \tilde{Y}_t^j, \tilde{Z}_t^{j-1}, \tilde{U}_t) dt + c^{j-1}(T-t, \tilde{X}_t) dt \\ &\quad + \tilde{Z}_t^{j-1} d\tilde{W}_t, \end{aligned} \quad (17)$$

where $\tilde{Z}_t^{j-1} = \frac{\partial \phi^{j-1}}{\partial x}(T-t, \tilde{X}_t)g(T-t, \tilde{X}_t)$ and

$$c^{j-1}(t, x) := \frac{\partial \phi^{j-1}}{\partial x}(t, x)s(t, x) + \text{Tr}(G(t, x) \frac{\partial^2 \phi^{j-1}}{\partial x^2}(t, x)).$$

The simulated trajectories are then used to update ϕ^j via the regression

$$\phi^j := \arg \min_{\psi \in \mathcal{G}} \mathbb{E} \left[\int_0^T \|\tilde{Y}_{T-t}^j - \psi(t, \tilde{X}_{T-t})\|^2 dt \right], \quad (18)$$

where $\mathcal{G}$ is a user-specified function class. In practice, the expectation is approximated by simulating N independent trajectories of $(\tilde{X}_t, \tilde{Y}_t^j)$ via Euler–Maruyama discretization of (11)–(17).

For the convergence analysis, we set aside discretization and optimization errors and assume both the SDE (17) and the minimization (18) are solved exactly. Define the operators

$$\begin{aligned} \mathcal{A}_t \phi &:= \frac{\partial \phi}{\partial x}(f + gk - s) - \frac{1}{2} \text{Tr}(G \frac{\partial^2 \phi}{\partial x^2}) + \frac{\partial f}{\partial x}^\top \phi, \\ \mathcal{A}_t' \phi &:= \frac{\partial \phi}{\partial x}s + \text{Tr}(G \frac{\partial^2 \phi}{\partial x^2}), \end{aligned}$$

and the projection onto $\mathcal{G}$:

$$\Pi^{\mathcal{G}}(\phi) := \arg \min_{\psi \in \mathcal{G}} \mathbb{E} \left[\int_0^T \|\phi(t, \tilde{X}_{T-t}) - \psi(t, \tilde{X}_{T-t})\|^2 dt \right].$$

Proposition 1: Assume $g(t, x) = g(t)$. Then the j -th iterate satisfies

$$\tilde{Y}_t = \psi^j(T-t, \tilde{X}_t) + M_t, \quad \phi^j = \Pi^{\mathcal{G}}(\psi^j) \quad (19)$$

where M_t is a martingale with respect to the filtration generated by $\tilde{W}_t$, and ψ^j solves the PDE

$$\frac{\partial \psi^j}{\partial t} + \mathcal{A}_t \psi^j + \mathcal{A}_t' \phi^{j-1} = 0, \quad \psi^j(T, x) = \nabla \ell_f(x). \quad (20)$$

Proof: Applying Itô's formula to $\psi^j(T-t, \tilde{X}_t)$ and using the PDE (20) gives

$$\begin{aligned} d\psi^j(T-t, \tilde{X}_t) &= \mathcal{A}_t' \phi^{j-1}(T-t, \tilde{X}_t) dt + \frac{\partial f}{\partial x}^\top \psi^j(T-t, \tilde{X}_t) \\ &\quad + \frac{\partial \psi^j}{\partial x} g(T-t, \tilde{X}_t) d\tilde{W}_t. \end{aligned}$$

Subtracting from (17) and using $c^{j-1}(t, x) = \mathcal{A}_t' \phi^{j-1}(t, x)$ and $\frac{\partial H}{\partial x}(T-t, \tilde{X}_t, \tilde{Y}_t^j, \tilde{Z}_t^{j-1}, \tilde{U}_t) = \frac{\partial f}{\partial x}^\top(T-t, \tilde{X}_t)\tilde{Y}_t$ shows that the difference $\tilde{Y}_t^j - \psi^j(T-t, \tilde{X}_t)$ satisfies

$$d[\tilde{Y}_t^j - \psi^j(T-t, \tilde{X}_t)] = \frac{\partial f}{\partial x}^\top [\tilde{Y}_t - \psi^j(T-t, \tilde{X}_t)] + d\tilde{M}_t$$

where $d\tilde{M}_t = \tilde{Z}_t^{j-1} d\tilde{W}_t - \frac{\partial \psi^j}{\partial x} g(T-t, \tilde{X}_t) d\tilde{W}_t$. Since $\tilde{Y}_0^j - \psi^j(T, \tilde{X}_0) = \nabla \ell_f(\tilde{X}_0) - \nabla \ell_f(\tilde{X}_0) = 0$, the difference $\tilde{Y}_t^j - \psi^j(T-t, \tilde{X}_t)$ is a Martingale concluding (19). Substituting into (18) gives the expression for ϕ^j . ■

Our next goal is concerned with the convergence of the iterates ϕ^j . Let ψ^* denote the fixed point satisfying

$$\frac{\partial \psi^*}{\partial t} + \mathcal{A}_t \psi^* + \mathcal{A}_t' \phi^* = 0, \quad \psi^*(T, x) = \nabla \ell_f(x)$$

where $\phi^* = \Pi^{\mathcal{G}}(\psi^*)$. When $\Pi_{\mathcal{G}}$ is the identity, $\psi^* = \phi^*$ recovers the solution to PDE (8).

To state the convergence result, we specialize to the case where $\mathcal{G}$ is a finite-dimensional function class. Specifically, let $\mathcal{H} := L^2([0, T] \times \mathbb{R}^n; \mathbb{R}^n)$ be equipped with the inner product

$$\langle \phi, \psi \rangle_{\mathcal{H}} := \int_0^T \int_{\mathbb{R}^n} \phi(x)^\top \psi(x) \rho_t(x) dx dt,$$

where ρ_t is the marginal density of $\tilde{X}_t$. Assume $\mathcal{G} = \text{span}\{\phi_1, \dots, \phi_K\}$, where $\{\phi_k\}_{k=1}^K \subset \mathcal{H}$ are orthonormal basis functions. Under this assumption, the projection $\Pi^{\mathcal{G}} : \mathcal{H} \rightarrow \mathcal{G}$ is the orthogonal projection $\Pi_{\mathcal{G}}(\phi) = \sum_{k=1}^K \langle \phi, \phi_k \rangle_{\mathcal{H}} \phi_k$, which is a bounded linear operator

on $\mathcal{H}$. For any $t \in [0, T]$ define the norm $\|\phi\|_{\mathcal{H}_t}^2 := \int \|\phi(x)\|^2 \rho(t, x) dx$. Define the constant

$$M := \sup_{t, s \in [0, T]} \left\| \Pi_t^{\mathcal{G}} \mathcal{P}_{t,s} \mathcal{A}_s' \Pi_s^{\mathcal{G}} \right\|_{\text{op}}, \quad (21)$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm induced by $\|\cdot\|_{\mathcal{H}_s} \rightarrow \|\cdot\|_{\mathcal{H}_t}$, $\Pi_t^{\mathcal{G}}(\phi) := \Pi^{\mathcal{G}}(\phi)(t, \cdot)$ is the evaluation of the projection at time t , and $\mathcal{P}_{t,s}$ is the semigroup associated with $\mathcal{A}_t$. Since $\mathcal{G}$ is finite-dimensional, the composition is represented, in the basis $\{\phi_k\}$, by the $K \times K$ matrix. Under smoothness assumptions on the basis functions, each matrix entry is finite and $M < \infty$.

Proposition 2: Under the above assumptions, the iterates ϕ^j converge to the fixed point ϕ^* at a superlinear rate:

$$\sup_{t \in [0, T]} \|\phi^j(t, \cdot) - \phi^*(t, \cdot)\|_{\mathcal{H}_t} \leq \frac{(MT)^j}{j!} \sup_{t \in [0, T]} \|\phi^0(t, \cdot) - \phi^*(t, \cdot)\|_{\mathcal{H}_s}.$$

In particular, $\phi^j \rightarrow \phi^*$ as $j \rightarrow \infty$ for any initialization ϕ^0 .

Proof: The proof follows a Picard iteration argument. By linearity of $\Pi_{\mathcal{G}}$, the error $e^j = \psi^j - \psi^*$ solves

$$\frac{\partial e^j}{\partial t} + \mathcal{A}_t e^j + \mathcal{A}_t' \Pi_t^{\mathcal{G}}(e^{j-1}) = 0, \quad e^j(T, x) = 0, \quad (22)$$

which yields

$$e^j(t, \cdot) = \int_0^t \mathcal{P}_{t,s} \mathcal{A}_s' \Pi_s^{\mathcal{G}}(e^{j-1})(s, \cdot) ds.$$

Setting $\tilde{e}^j := \Pi^{\mathcal{G}}(e^j) = \phi^j - \phi^*$ and $\mathcal{L}_{t,s} := \Pi_t^{\mathcal{G}} \mathcal{P}_{t,s} \mathcal{A}_s' \Pi_s^{\mathcal{G}}$,

$$\tilde{e}^j(t, \cdot) = \int_0^t \mathcal{L}_{t,s} \tilde{e}^{j-1}(s, \cdot) ds.$$

Iterating it from $j = J$ to $j = 1$ concludes

$$\tilde{e}^J(s_J, \cdot) = \int_0^{s_J} \dots \int_0^{s_1} \mathcal{L}_{s_J, s_{J-1}} \dots \mathcal{L}_{s_2, s_1} \tilde{e}^0(s_1, \cdot) ds_1 \dots ds_J.$$

Taking the norm of both sides and applying (21) yields

$$\|\tilde{e}^J(t, \cdot)\| \leq \frac{(MT)^J}{J!} \sup_s \|\tilde{e}^0(s, \cdot)\|, \quad \forall t \in [0, T],$$

concluding the proof. $\blacksquare$

We note that there remains a gap between this theoretical result and the practical algorithm, where $\mathcal{G}$ is parameterized by a neural network and the minimization (18) is solved approximately; closing this gap is left for future work.

V. NUMERICAL EXPERIMENTS

In this section, we present numerical results for four experiments: a linear model, a stochastic pendulum, the full fine-tuning procedure on a toy diffusion model, and a double-well benchmark. The first two experiments are designed to assess the ability of the proposed framework to compute the adapted adjoint process and to compare its accuracy, for gradient estimation, against the non-adapted adjoint process. The third and fourth experiments compare the performance of the proposed TR-BSDE method against adjoint matching, which relies on the non-adapted adjoint process. The code for reproducing the results and the details of the numerical procedures are available online ¹.

Algorithm 2 Fine-tuning the diffusion model

- 1: **Input:** Iteration number k_f , optimization iteration number o_f , initial distribution q_0 with parameters μ and Q , initial control law $k(t, x) = 0$.
 - 2: **for** $k \in \{1, 2, \dots, k_f\}$ **do**
 - 3: Obtain the function ϕ from Algorithm 1
 - 4: Update the feedback control law according to $k(t, x) = -g(t)^\top \phi(t, x)$
 - 5: **if** $k \bmod 5 = 0$ **then**
 - 6: Update μ and Q according to (28) using the ADAM optimizer for o_f iterations
 - 7: **end if**
 - 8: **end for**
 - 9: **Output:** Feedback control law $k(t, x)$, optimal initial distribution q_0
-

Algorithm 3 Projected non-adapted adjoint process

- 1: **Input:** sample size N , step-size Δt , iteration J_f , function class Φ , initial distribution q_0 .
 - 2: Initialize $\{X_0^i\}_{i=1}^N \sim q_0$
 - 3: Obtain $\{X_t^i\}_{i=1}^N$ by simulating (2)
 - 4: Initialize function $\phi \in \Phi$
 - 5: $Y_T^i = \nabla \ell_f(X_T^i)$
 - 6: Obtain $\{Y_t^i\}_{i=1}^N$ by simulating (4)
 - 7: Update ϕ using ADAM optimizer minimizing (25)
 - 8: **Output:** $\{\phi(t, \cdot)\}_{t \in \{0, \Delta t, \dots, T\}}$
-

Algorithm 4 Adjoint matching fine-tuning

- 1: **Input:** Iteration number k_f , control input iteration number J_f , optimization iteration number o_f , function class Φ , initial distribution q_0 with parameters μ and Q , initial control law $k(t, x) = 0$.
 - 2: **for** $k \in \{1, 2, \dots, k_f\}$ **do**
 - 3: **for** $j \in \{1, 2, \dots, J_f\}$ **do**
 - 4: Obtain $\{X_t^i\}_{i=1}^N$ by simulating (2)
 - 5: $Y_T^i = \nabla \ell_f(X_T^i)$
 - 6: Obtain $\{Y_t^i\}_{i=1}^N$ by simulating (4)
 - 7: Update control law $k(t, x)$ using ADAM optimizer minimizing $\frac{1}{N} \sum_{i=1}^N \|k(t, X_t^i) + g(t)^\top Y_t^i\|$
 - 8: **end for**
 - 9: **if** $k \bmod 5 = 0$ **then**
 - 10: Update μ and Q according to (28) with $\{Y_0^i\}_{i=1}^N$ using the ADAM optimizer for o_f iteration
 - 11: **end if**
 - 12: **end for**
 - 13: **Output:** Feedback control law $k(t, x)$, optimal initial distribution q_0
-

¹<https://github.com/YuhangMeiUW/BSDE-Gradient>

A. Verification in linear quadratic setting

We consider the 2-dimensional linear system

$$dX_t = AX_t dt + \varepsilon B dW_t, \quad X_0 = \xi \sim q_0, \quad (23)$$

and the objective function with the quadratic terminal cost

$$\mathbb{E}[J(\xi)] := \mathbb{E} \left[\frac{1}{2} X_T^\top Q_f X_T \right], \quad (24)$$

where $\varepsilon > 0$ denotes the strength of the noise,

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -0.5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad Q_f = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

q_0 is standard Gaussian, and $T = 2$. We simulate and compare two numerical methods to estimate the derivative $\frac{\partial \mathbb{E}[J(\xi)]}{\partial \xi}$. The ground-truth is $\phi(0, \xi)$ where ϕ solves the PDE (8). In the linear case, the exact solution is $\phi(t, x) = G(t)x$ where $G(t)$ solves the differential Riccati equation [23, Eq. 15].

Time-reversed BSDE (TR-BSDE): To estimate $\phi(0, x)$, we use the iterative procedure that alternates between simulating the time-reversed BSDE (15) and solving the optimization (16). Although the ground-truth function ϕ is linear in this example, we parameterize it with a neural network in order to validate the proposed training procedure. We perform 10 outer iterations. In each iteration, we simulate (15), then train the neural network for ϕ over 2000 optimization steps, and then use the updated ϕ in the next round of BSDE simulation, as described in Algorithm 1.

Projected non-adapted adjoint process (PNAA): To estimate $\phi(0, x)$, we use the relation $\phi(0, \xi) = Y_0 = \mathbb{E}[Y_0 | \mathcal{F}_0]$, where Y denotes the non-adapted adjoint process. Specifically, we simulate N sample trajectories $\{(X^i, Y^i)\}_{i=1}^N$ with random initial conditions $X_0^i = \xi^i \sim q_0$, and then estimate ϕ by solving the regression problem

$$\min_{\phi \in \Phi} \sum_{i=1}^N \|Y_t^i - \phi(t, X_t^i)\|^2. \quad (25)$$

The procedure is detailed in Algorithm 3. To ensure a fair comparison with TR-BSDE, we use the same neural network architecture and the same total number of training iterations.

To quantify the accuracy of gradient estimation, we use the following mean-squared error (MSE) criterion:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N \|G(0)\xi^i - \phi(0, \xi^i)\|_2^2, \quad \xi^i \sim q_0, \quad (26)$$

where $G_0 \xi^i$ is the ground truth value and $\phi(0, \xi)$ is its estimate obtained from the two methods above. Figure 1a illustrates the effect of the noise strength ε on the performance of the two estimators. As expected, the MSE increases with ε for both methods. However, the proposed TR-BSDE method consistently achieves lower error than PNAA across all noise levels, reflecting the fact that the non-adapted adjoint process yields inherently noisier gradient estimates.

B. Optimal initial distribution for inverted pendulum

We consider the nonlinear stochastic pendulum

$$f(x) = \begin{bmatrix} x(2) \\ \sin(x(1)) - 0.01x(2) \end{bmatrix}, \quad g(x) = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix},$$

and the objective

$$\mathbb{E}[J(\xi)] := \mathbb{E} \left[\frac{1}{2} X_T(2)^2 + 1 - \cos(X_T(1)) \right], \quad \xi \sim q_0. \quad (27)$$

We study the problem of optimizing $\mathbb{E}[J(\xi)]$ with respect to the initial distribution q_0 , assumed to have the empirical form

$$q_0 = \frac{1}{N} \sum_i^N \delta_{\theta^i},$$

where the support points θ^i are treated as optimization variables. The gradient of the objective with respect to each variable θ^i is given by $\phi(0, \theta^i)$, where ϕ solves the PDE (8).

We approximate ϕ using both TR-BSDE (Algorithm 1) and PNAA (Algorithm 3), and use the resulting gradients to optimize for θ^i through a gradient descent procedure. For comparison, we also implement a standard automatic differentiation (AutoDiff) baseline, which computes gradients by backpropagating through a time-discretized version of the SDE. We run the optimization algorithm for 15000 iterations. The resulting accumulation points of θ^i are shown in Fig 1b and Fig. 1c, together with a heat map of the objective function $\mathbb{E}[J(\xi) | \xi = \theta]$, estimated by Monte Carlo simulation. It is observed that the points produced by TR-BSDE concentrate more tightly in low-cost regions than those obtained by PNAA. Moreover, the PNAA result includes several points that drift far from the origin and are therefore omitted from the figure. Compared with the AutoDiff baseline, TR-BSDE also converges substantially faster. Overall, these results indicate that the proposed method yields more accurate and lower-variance gradient estimates for this nonlinear system.

C. Fine-tuning diffusion model

We consider a pretrained one-dimensional diffusion model that generates samples from the bimodal distribution

$$p_T = 0.5 \mathcal{N}(-3, 1) + 0.5 \mathcal{N}(3, 1)$$

Our goal is to fine-tune this model so that it generates samples from the tilted distribution defined through the terminal cost

$$\ell_f(x) = \beta \frac{1}{2} (x-3)^2$$

where $\beta > 0$ is the tilting parameter. We assume the nominal initial distribution $p_0 = \mathcal{N}(0, 1)$, and parameterize the modified initial distribution $q_0 = \mathcal{N}(\mu, Q^2)$, where μ and Q are optimization variables. The objective is to solve the SOC problem (3) by jointly optimizing the initial distribution q_0 and the control law $U_t = k(t, X_t)$.

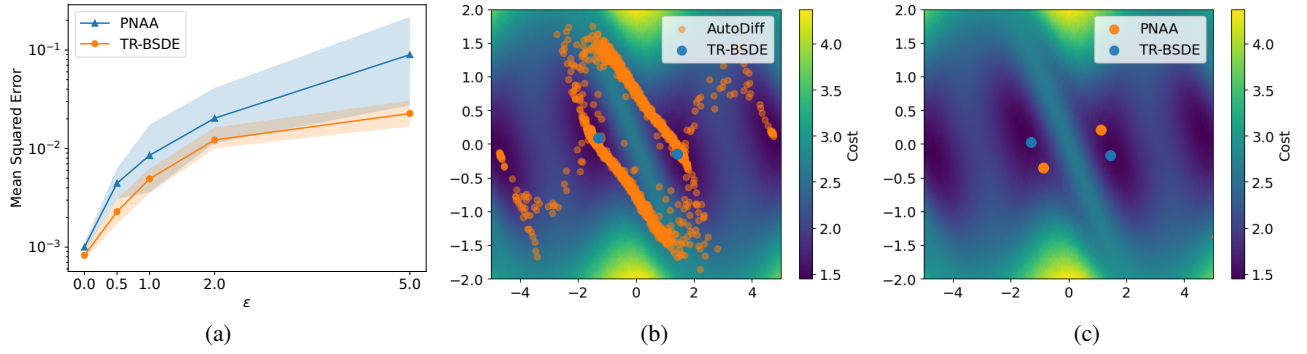


Fig. 1: Numerical results for comparing the accuracy of gradient estimation methods. (a) influence of noise strength ε on MSE for the linear system in Sec. V-A. The solid line represents the mean, and the shaded region represents the range from the minimum to the maximum across 10 experiments. (b-c) Optimal initial distribution for the nonlinear system in Sec. V-B. The heat map represents the optimization objective function and the points represent the support of the empirical distribution.

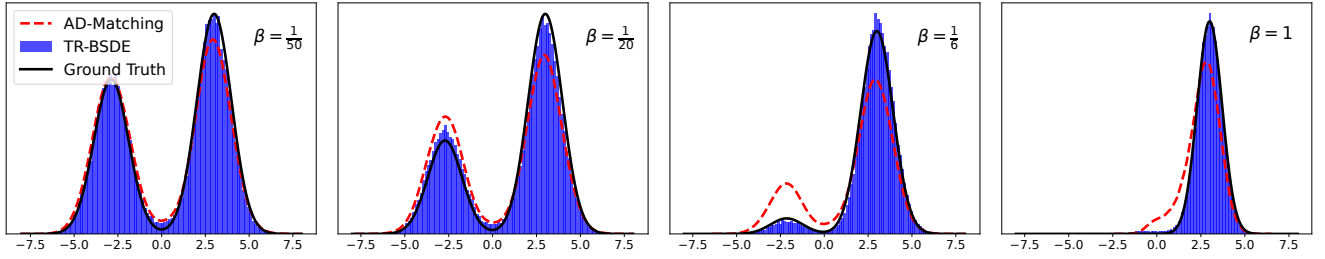


Fig. 2: Numerical result for comparing TR-BSDE and adjoint matching on fine-tuning the diffusion model (Sec. V-C): The black solid line denotes the target tilted distribution, the red dashed line denotes the fine-tuned result from the adjoint matching method, and the blue histogram denotes the fine-tuned result from TR-BSDE method.

Initial distribution optimization: To optimize the initial distribution, we use the closed-form expression for the KL divergence between one-dimensional Gaussian distributions

$$D_{\text{KL}}(q_0 \| p_0) = \frac{1}{2}(Q^2 + \mu^2 - \log Q^2 - 1),$$

then we can obtain the the gradient of cost with respect to the initial distribution mean and variance

$$\frac{\partial}{\partial \mu} [\mathbb{E}_{\xi \sim q_0} [J(\xi, U)] + D_{\text{KL}}(q_0 \| p_0)] = \mathbb{E}[Y_0] + \mu, \quad (28a)$$

$$\frac{\partial}{\partial Q} [\mathbb{E}_{\xi \sim q_0} [J(\xi, U)] + D_{\text{KL}}(q_0 \| p_0)] = \mathbb{E}[Y_0(X_0 - \mu)/Q] + Q - 1/Q. \quad (28b)$$

TR-BSDE: To compute the optimal control law, we initialize with the zero control $k(t, x) = 0$, solve the BSDE, and learn ϕ from Algorithm 1. After each update of ϕ , we set $k(t, x) = -g(t)^\top \phi(t, x)$. This choice is motivated by the stochastic maximum principle, which implies that at optimality $U_t = -g(t)^\top Y_t$ and $Y_t = \phi(t, X_t)$. The parameters of the initial Gaussian distribution are updated using (28) and $Y_0 = \phi(0, x)$. The procedure appears in Algorithm 2.

Adjoint matching [12]: The control law $k(t, x)$ is parameterized by a neural network and learned by regressing $k(t, X_t)$ onto $-g(t)^\top Y_t$, where Y_t is the non-adapted adjoint process.

After a fixed number of regression iterations, the adjoint process is re-simulated under the updated control law. The procedure appears in Algorithm 4. As in TR-BSDE, the parameters of the initial Gaussian distribution are updated using (28), with the difference that Y_0 is replaced by the non-adapted process Y_0 .

We compare two methods for tilting parameter $\beta \in \{\frac{1}{50}, \frac{1}{20}, \frac{1}{8}, \frac{1}{6}, 1\}$. The experimental results are shown in Fig. 2. It is observed that when β is small, the two methods exhibit similar performance. However, as β increases, the fine-tuning task becomes more challenging, and the proposed TR-BSDE outperforms the Adjoint matching approach.

D. Double-well benchmark

We evaluate TR-BSDE and adjoint matching on the double-well benchmark from [20] as the problem dimension increases. The model is governed by the SDE

$$dX_t = -\nabla \Psi(X_t) dt + U_t dt + dW_t, \quad X_0 = 0, \quad (29)$$

with cost

$$J = \mathbb{E} \left[\int_0^T \frac{1}{2} \|U_s\|^2 ds + \ell_f(X_T) \right], \quad (30)$$

where

$$\Psi(x) = \sum_{i=1}^n \kappa_i (x_i^2 - 1)^2, \quad \ell_f(x) = \sum_{i=1}^n v_i (x_i^2 - 1)^2.$$

We set $\kappa_i = v_i = 1$, $T = 1$, $\Delta t = 0.005$, and vary the dimension $n \in \{5, 6, 7, 8, 9, 10\}$. Performance is measured by the relative cost gap $(J - J^*)/J^*$, estimated with $N = 10000$ samples, where J^* is the optimal cost. The uncontrolled system is included as a baseline. As shown in Fig. 3, TRBSDE consistently outperforms adjoint matching across all dimensions, with the relative cost gap remaining small and stable as n increases

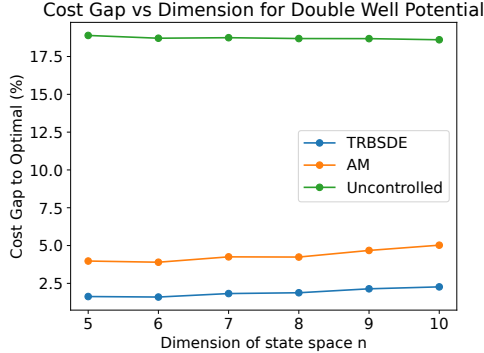


Fig. 3: Comparison of TRBSDE and Adjoint Matching on the double well benchmark across state dimensions $n \in \{5, 6, 7, 8, 9, 10\}$. The y-axis reports the relative cost gap $(J - J^*)/J^*$, where J is the realized cost under the learned controller and J^* is the optimal cost obtained from the path integral representation. Each estimate uses $N = 10000$ Monte Carlo samples. TRBSDE consistently achieves a smaller gap than Adjoint Matching as n increases.

Remark 2 (Computational limitations): The main computational bottleneck of TRBSDE, relative to adjoint matching, is the trace term in (15), which requires evaluating the Hessian of a neural network and becomes increasingly expensive as n grows. Adjoint matching avoids this cost but incurs higher gradient variance, as demonstrated both theoretically in Section III and numerically. A natural remedy is to replace the exact trace with Hutchinson’s estimator [36], a stochastic approximation that has proven effective in similar settings [37], and which we leave for future work. Additionally, TRBSDE requires an estimated score function, which is an extra ingredient compared to adjoint matching.

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